[10/28/2026 10:09 PM] Nakama :

ESHRAGH UNIVERSITY

Department of Computer Engineering

EMBEDDED SYSTEMS

Longest Remaining Time First (LRTF) Scheduling

Theory, Semaphore & Monitor Implementation, and Comparative Analysis with EDF and RM

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Abstract

This article presents a comprehensive study of the **Longest Remaining Time First (LRTF)** CPU scheduling algorithm, covering its theoretical foundations, mathematical formulation, pseudocode, and practical implementation using both **semaphores** and **monitors**. LRTF is the preemptive variant of Longest Job First (LJF) and selects the process with the largest remaining burst time at every scheduling point. While it guarantees that all processes complete at approximately the same time and is often considered starvation-free under static workloads, it suffers from high average waiting time and turnaround time, and can induce starvation under dynamic arrivals.

The article provides **18 tables** and **18 figures**, including colored flowcharts, Gantt charts, architecture diagrams, and performance comparisons. We implement LRTF using semaphores, then using monitors, and finally present a **hybrid semaphore–monitor implementation**. A comparative analysis against **Earliest Deadline First (EDF)** and **Rate Monotonic (RM)** scheduling is conducted using simulation data. The article also includes a real-thread tutorial, operating-system timing comparisons, and a final critical analysis with references.

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1. Introduction to Real-Time CPU Scheduling

CPU scheduling is the decision-making process that determines which ready task executes next on a processor. In real-time systems, scheduling is not merely a performance optimization; it is a **correctness requirement**. A scheduling algorithm must ensure that every task completes before its deadline, or at least that the fraction of missed deadlines remains bounded and acceptable.

Classical real-time scheduling algorithms fall into two broad categories: **fixed-priority** algorithms such as Rate Monotonic (RM), and **dynamic-priority** algorithms such as Earliest Deadline First (EDF) and Least Laxity First (LLF). LRTF belongs to a third, less conventional category: **execution-time-driven** scheduling, where the priority of a job is a function of its remaining execution time rather than its deadline or period.

LRTF is the preemptive version of Longest Job First (LJF). At each scheduling instant, the scheduler inspects all ready jobs and dispatches the one with the **largest remaining burst time**. If a new job arrives with a remaining time larger than that of the currently running job, preemption occurs. This greedy strategy has an interesting property: if all jobs arrive simultaneously, they all finish at approximately the same time. However, this property comes at the cost of very poor average waiting time and response time for short jobs.

> **Scope of this article.** We focus on uniprocessor preemptive scheduling of independent jobs. Extensions to multiprocessor systems, precedence constraints, and resource sharing are discussed only where relevant to the implementation sections.

1.1 Why Study LRTF?

LRTF is pedagogically important because it exposes the trade-off between **fairness in completion time** and **responsiveness**. It is also a useful baseline for evaluating more sophisticated algorithms. In cloud computing and batch processing, LRTF-like policies have been used as a comparison point to demonstrate the inefficiency of prioritizing long jobs

when user-perceived latency matters. Understanding LRTF deeply therefore provides insight into why EDF and RM dominate practical real-time systems.

2. LRTF Fundamentals

2.1 Definition

Longest Remaining Time First (LRTF) is a preemptive scheduling discipline in which the processor is always assigned to the ready job with the maximum remaining processing time. Formally, let $R_i(t)$ denote the remaining execution time of job i at time t. The scheduler selects job j such that: [10/28/2026 10:09 PM] Nakama :

$$j = \arg\max_{i \in \mathcal{R}(t)} R_i(t)$$

where $\mathcal{R}(t)$ is the set of ready jobs at time t. Ties are broken by arrival time (FCFS) or by process ID, depending on the implementation.

2.2 Relationship to LJF and SRTF

LRTF is the preemptive counterpart of **Longest Job First (LJF)**, just as **Shortest Remaining Time First (SRTF)** is the preemptive counterpart of Shortest Job First (SJF). The four algorithms form a 2×2 matrix along two axes: preemptive vs. non-preemptive, and longest vs. shortest. LRTF occupies the (preemptive, longest) cell. This relationship is summarized in Table 1 and visualized in Figure 1.

Table 1: Matrix of Job-Selection Algorithms

	Shortest First	Longest First
Non-preemptive	SJF (Shortest Job First)	LJF (Longest Job First)
Preemptive	SRTF (Shortest Remaining Time First)	**LRTF (Longest Remaining Time First)**

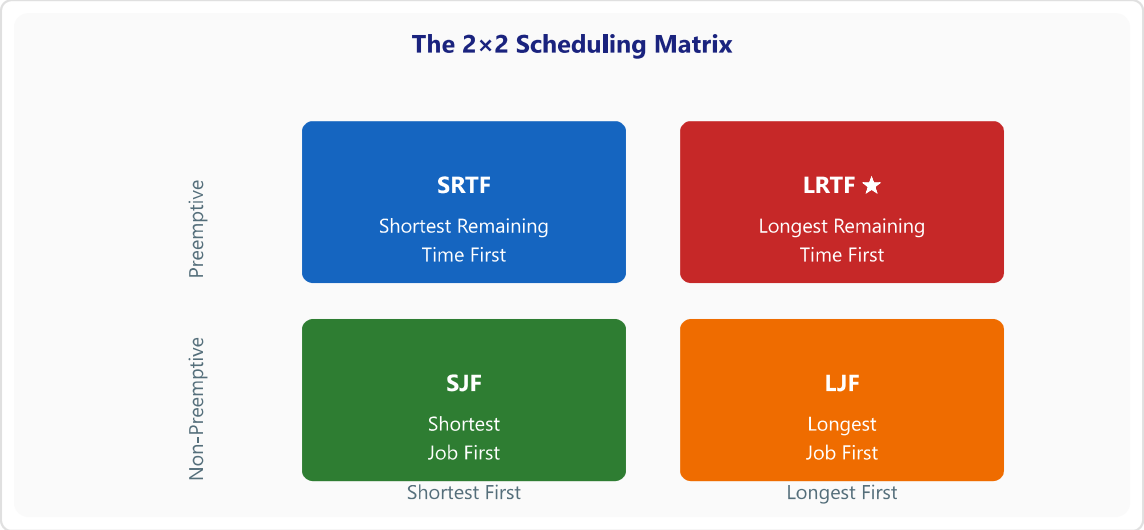


Figure 0: *Visualization of the 2×2 matrix of job-selection algorithms; LRTF is highlighted.*

2.3 Key Characteristics

- **Preemptive:** A running job can be interrupted if a newly arrived job has a larger remaining time.
- **Execution-time-driven priority:** Priority is inversely related to the remaining burst time (larger remaining time ⇒ higher priority).
- **Completion-time equalization:** If all jobs are available at $t = 0$, they tend to finish at approximately the same time.
- **High context-switch frequency:** The scheduler must re-evaluate the maximum remaining time at every arrival or completion event, and often at every time quantum.
- **Poor short-job responsiveness:** Short jobs can be repeatedly preempted by long jobs, leading to very high waiting times.

2.4 Process State View

Under LRTF, a process traverses a state space that includes *new*, *ready*, *running*, *preempted*, *waiting*, and *terminated*. The *preempted* state, shown in Figure 2, is particularly frequent because any arrival with a larger remaining time forces a context switch.

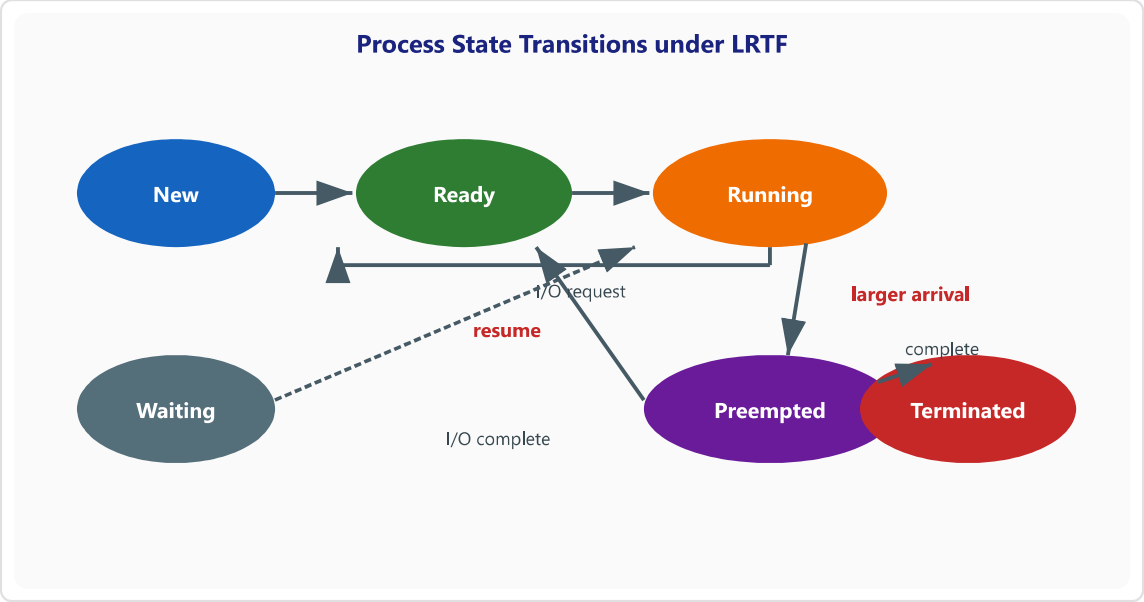


Figure 0: *Process state transitions under LRTF. Preemption by larger arrivals is frequent.*

2.5 Comparison of Basic Properties

Table 2: Qualitative Comparison of LRTF with SRTF, LJF, and SJF

Property	LRTF	SRTF	LJF	SJF
Preemptive	Yes	Yes	No	No
Optimizes average waiting time	No (worst-case)	Yes (optimal)	No	Yes (optimal)
Starvation risk	Low (static), high (dynamic)	High for long jobs	High for short jobs	High for long jobs
Convoy effect	Yes (long-job convoy)	No (short jobs first)	Yes	No
Context switches	High	High	Low	Low

2.6 Decision Tree: SRTF vs. LRTF

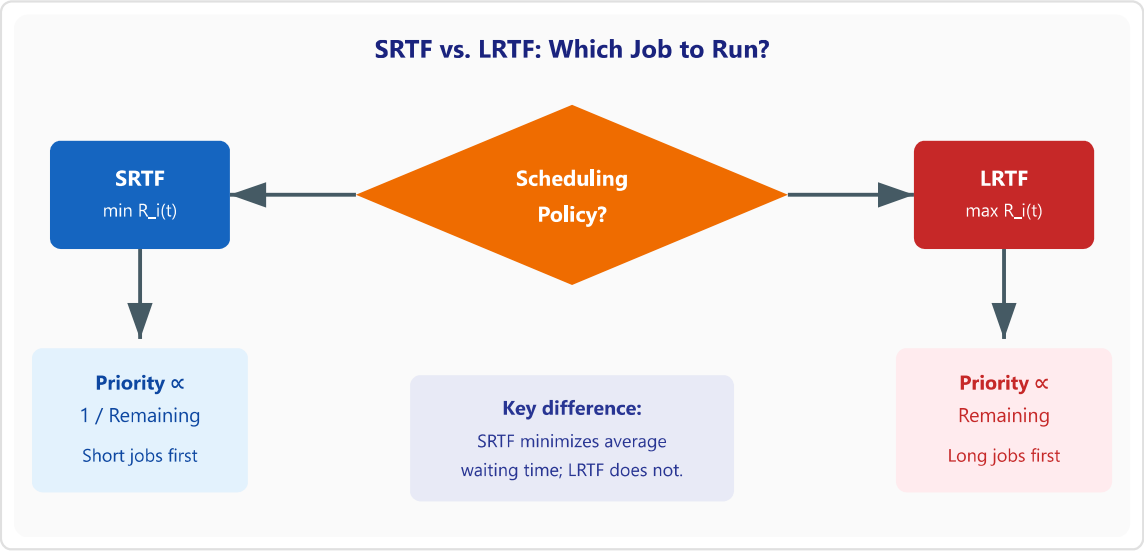


Figure 0: *Decision tree contrasting SRTF and LRTF selection rules.*

3. Pseudocode and Flowchart

3.1 High-Level Pseudocode

The pseudocode below assumes a discrete time model in which the scheduler makes a decision at every integer time unit. In practice, decisions are made at arrival and completion events, but the unit-step model simplifies exposition and is used in most textbooks.

```
Algorithm LRTF_Scheduler(processes P)
Input:  Set of processes P = {p1, p2, ..., pn}
        Each pi has arrival time AT[i] and burst time BT[i]
Output: Schedule and metrics (CT, TAT, WT)

1:  currentTime ← 0
2:  remainingBT[i] ← BT[i] for all i
3:  completed ← 0
4:  readyQueue ← ∅

5:  while completed < n do
6:      for each pi with AT[i] == currentTime do
7:          add pi to readyQueue
8:      end for

9:      if readyQueue is empty then
10:         currentTime ← currentTime + 1
11:         continue
12:     end if

13:     // Select job with maximum remaining burst time
14:     selected ← argmax_{pi ∈ readyQueue} remainingBT[i]

15:     // Tie-breaking: FCFS (smallest arrival time), then smallest ID
16:     if tie then
17:         selected ← tieBreakFCFS(selected)
18:     end if

19:     // Execute selected for one time unit
20:     remainingBT[selected] ← remainingBT[selected] – 1
21:     currentTime ← currentTime + 1

22:     // Check for completion
23:     if remainingBT[selected] == 0 then
24:         CT[selected] ← currentTime
25:         TAT[selected] ← CT[selected] – AT[selected]
26:         WT[selected]  ← TAT[selected] – BT[selected]
27:         remove selected from readyQueue
28:         completed ← completed + 1
29:     end if
30: end while

31: return schedule, average TAT, average WT
```

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3.2 Colored Flowchart

Figure 4 presents the decision logic of LRTF as a colored flowchart. Green nodes represent initialization and termination, blue nodes represent data operations, orange nodes represent decisions and actions, and red nodes represent final states.

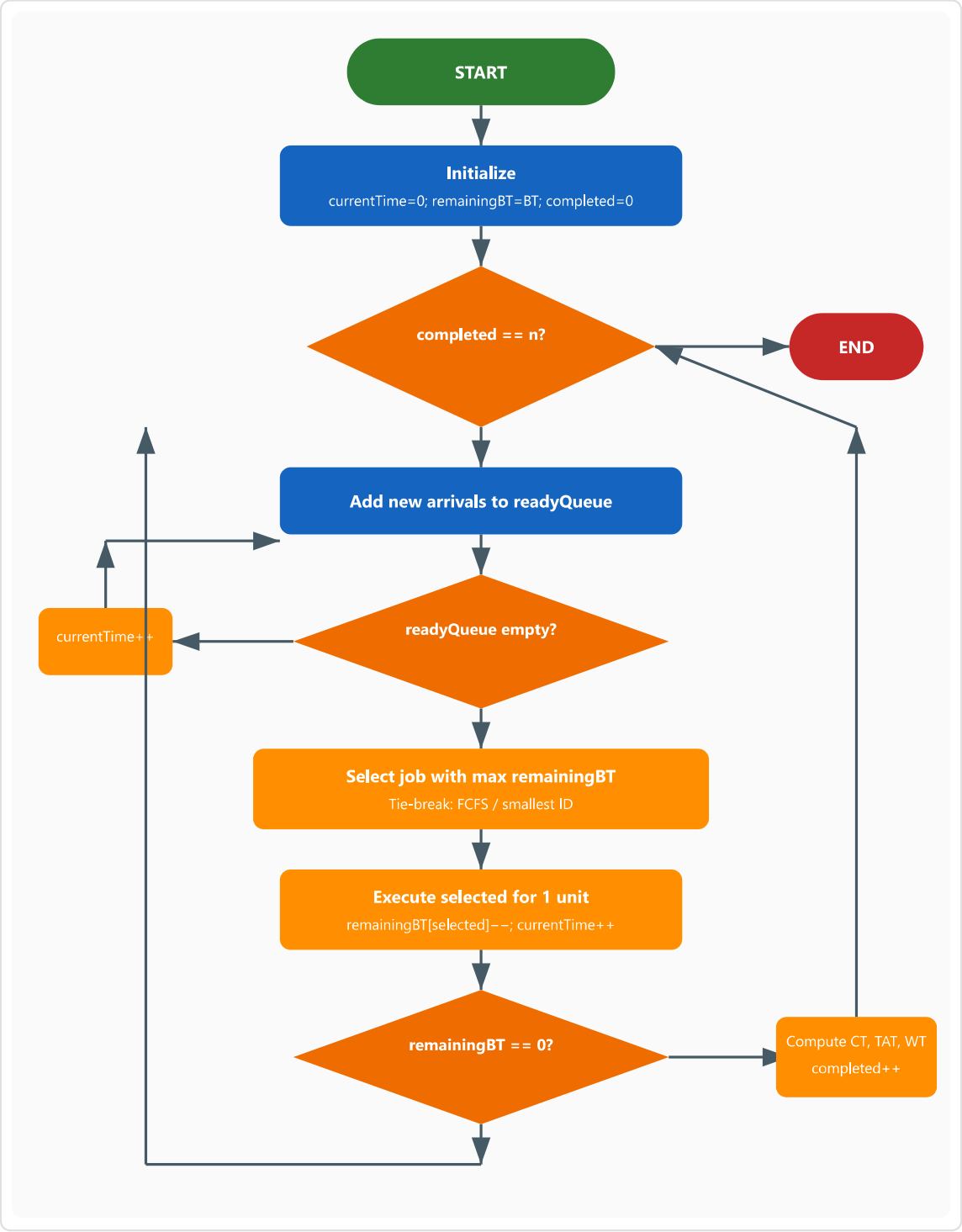


Figure 0: *Colored flowchart of the LRTF scheduling algorithm.*

4. Mathematical Formulation

4.1 Basic Metrics

For a process P_i with arrival time A_i, burst time B_i, and completion time C_i, the standard scheduling metrics are defined as follows:

$$\text{Turnaround Time: } TAT_i = C_i - A_i$$

$$\text{Waiting Time: } W_i = TAT_i - B_i = C_i - A_i - B_i$$

$$\text{Response Time: } R_i = \text{firstDispatchTime}_i - A_i$$

The averages are computed over n processes:

$$\overline{TAT} = \frac{1}{n}\sum_{i=1}^{n} TAT_i, \quad \overline{W} = \frac{1}{n}\sum_{i=1}^{n} W_i$$

4.2 LRTF Selection Rule

At any scheduling instant t, let $R_i(t)$ be the remaining execution time of process i. The LRTF scheduler dispatches:

$$P_{\text{selected}} = \arg\max_{i \in \mathcal{R}(t)} R_i(t)$$

If multiple processes attain the maximum, the tie is resolved by the smallest arrival time (FCFS):

$$\text{tie-break: } \min A_i \text{ among } \{i : R_i(t) = R_{\max}(t)\}$$

4.3 CPU Utilization and Schedulability

For a set of periodic tasks with execution times C_i and periods T_i, the processor utilization is:

$$U = \sum_{i=1}^{n} \frac{C_i}{T_i}$$

LRTF does not have a simple utilization bound analogous to the RM bound $U \leq n(2^{1/n} - 1)$ or the EDF bound $U \leq 1$. Its schedulability depends on the specific arrival pattern and execution-time distribution. This is illustrated in Figure 5, which compares the safe utilization regions of RM, EDF, and LRTF.

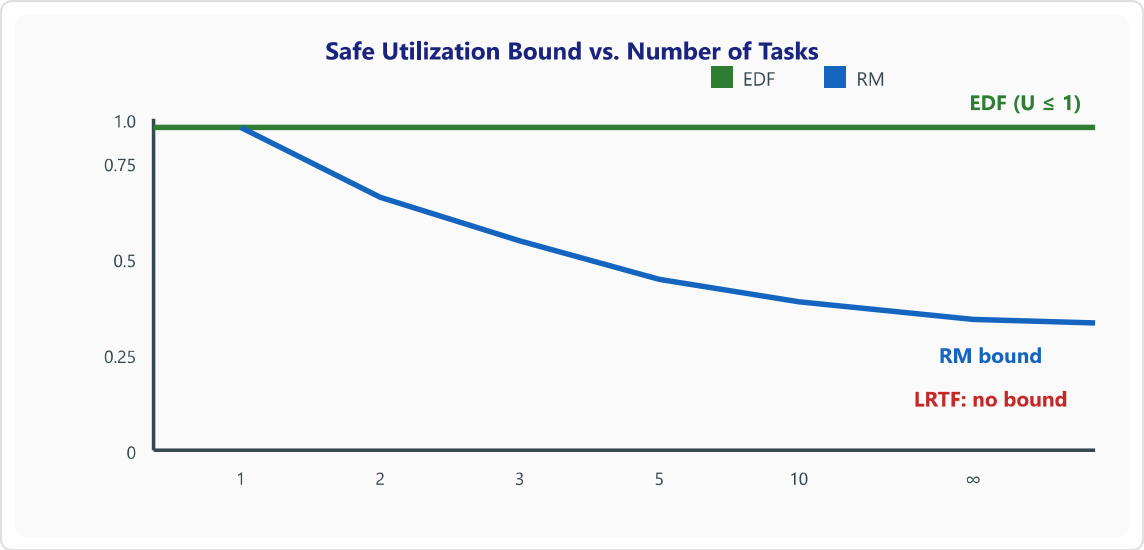


Figure 0: *Safe utilization bounds for EDF, RM, and LRTF. LRTF lacks a general bound.*

4.4 Complexity

At each decision point, LRTF must find the maximum remaining time among at most n ready jobs. Using a linear scan, the per-decision complexity is $O(n)$. If a binary heap is maintained, the complexity can be reduced to $O(\log n)$ for extraction and $O(\log n)$ for updates. However, LRTF may require a decision at every time unit in the worst case, leading to an overall scheduling complexity of $O(n \cdot L)$ for a schedule of length L, compared to $O(n \log n)$ for event-driven EDF.

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Table 3: Complexity Comparison of Scheduling Algorithms

Algorithm	Decision Points	Per-Decision Complexity	Overall (Worst Case)
FCFS	n	O(1)	O(n)
SJF / LJF	n	O(n)	$O(n^2)$
SRTF / LRTF	O(n·L)	O(n)	$O(n^2 \cdot L)$
RM (heap)	2n	O(log n)	O(n log n)
EDF (heap)	2n	O(log n)	O(n log n)

5. Worked Example with Gantt Chart

Consider four processes arriving at different times, as shown in Table 3. We trace the LRTF schedule step by step.

Table 4: Process Set for LRTF Example

Process	Arrival Time (ms)	Burst Time (ms)
P1	1	2
P2	2	4
P3	3	6
P4	4	8

5.1 Scheduling Trace

Table 5: Step-by-Step LRTF Scheduling Trace

Time	Ready Queue	Selected	Remaining After	Event
1	P1	P1	P1=1	—
2	P1(1), P2(4)	P2	P1=1, P2=3	P2 arrives
3	P1(1), P2(3), P3(6)	P3	P2=3, P3=5	P3 arrives
4	P1(1), P2(3), P3(5), P4(8)	P4	P3=5, P4=7	P4 arrives
5	P1(1), P2(3), P3(5), P4(7)	P4	P4=6	—
6	P1(1), P2(3), P3(5), P4(6)	P4	P4=5	—
7	P1(1), P2(3), P3(5), P4(5)	P3 / P4	Tie: FCFS → P3	Tie
8	P1(1), P2(3), P3(4), P4(5)	P4	P4=4	—
9	P1(1), P2(3), P3(4), P4(4)	P3 / P4	Tie → P3	Tie
10	P1(1), P2(3), P3(3), P4(4)	P4	P4=3	—
11	P1(1), P2(3), P3(3), P4(3)	P2	P2=2	3-way tie → FCFS
12	P1(1), P2(2), P3(3), P4(3)	P3 / P4	Tie → P3	Tie
13	P1(1), P2(2), P3(2), P4(3)	P4	P4=2	—
14	P1(1), P2(2), P3(2), P4(2)	P2	P2=1	4-way tie → FCFS
15	P1(1), P2(1), P3(2), P4(2)	P3 / P4	Tie → P3	Tie
16	P1(1), P2(1), P3(1), P4(2)	P4	P4=1	—
17	P1(1), P2(1), P3(1), P4(1)	P1	P1=0 ✓	P1 completes
18	P2(1), P3(1), P4(1)	P2	P2=0 ✓	P2 completes
19	P3(1), P4(1)	P3	P3=0 ✓	P3 completes
20	P4(1)	P4	P4=0 ✓	P4 completes

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5.2 Gantt Chart

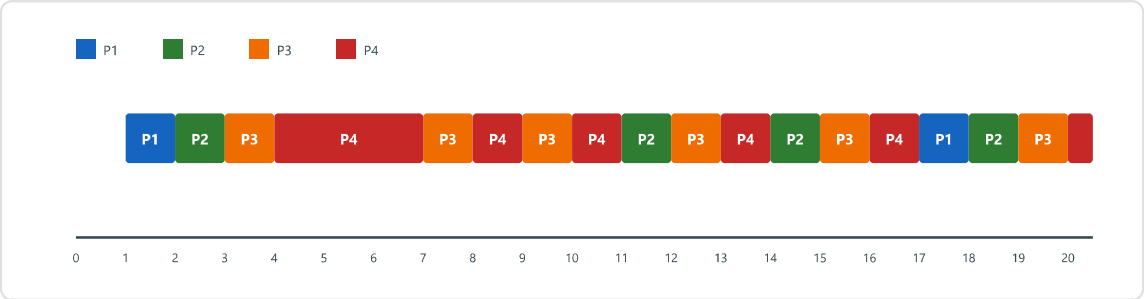


Figure 0: *Gantt chart for the LRTF schedule of Table 3. Preemptions are frequent because longer jobs repeatedly reclaim the CPU.*

5.3 Remaining-Time Evolution

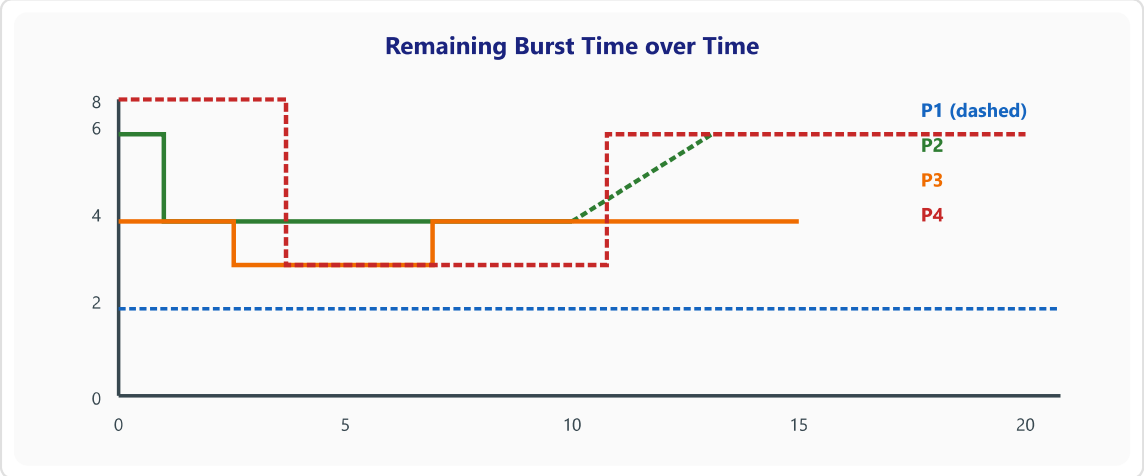


Figure 0: *Remaining-time evolution for the four processes. LRTF equalizes completion time, with all four finishing between t=17 and t=20.*

5.4 Computed Metrics

Table 6: Completion, Turnaround, and Waiting Times for the LRTF Example

Process	Arrival	Burst	Completion	TAT	Waiting
P1	1	2	18	17	15
P2	2	4	19	17	13
P3	3	6	20	17	11
P4	4	8	20	16	8
Average				**16.75**	**11.75**

The average waiting time of 11.75 ms is significantly higher than what SJF or SRTF would produce for the same workload. This illustrates the fundamental inefficiency of LRTF for average-case performance.

6. Semaphore-Based Implementation of LRTF

6.1 Role of Semaphores in Scheduling

A **semaphore** is a synchronization primitive consisting of an integer counter and a wait queue. The two atomic operations are $P(S)$ (wait/proberen) and $V(S)$ (signal/verhogen). In a scheduler, semaphores protect the ready queue and the shared scheduling state from concurrent access by multiple threads or interrupt handlers.

For LRTF, the critical shared data structures are:

- The **ready list** containing all runnable processes.
- The **remaining-time array** R_i.
- The **completion counter** and per-process metrics.

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Table 7: Semaphore Variables and Their Roles in LRTF

Semaphore	Initial Value	Purpose
`mutex_ready`	1	Protects the ready queue from concurrent insertion/removal.
`mutex_metrics`	1	Protects completion-time and waiting-time counters.
`sem_dispatch`	0	Signals the dispatcher that a scheduling decision is pending.
`sem_completion`	0	Signals the metrics thread that a process has completed.

6.2 Scheduler Architecture

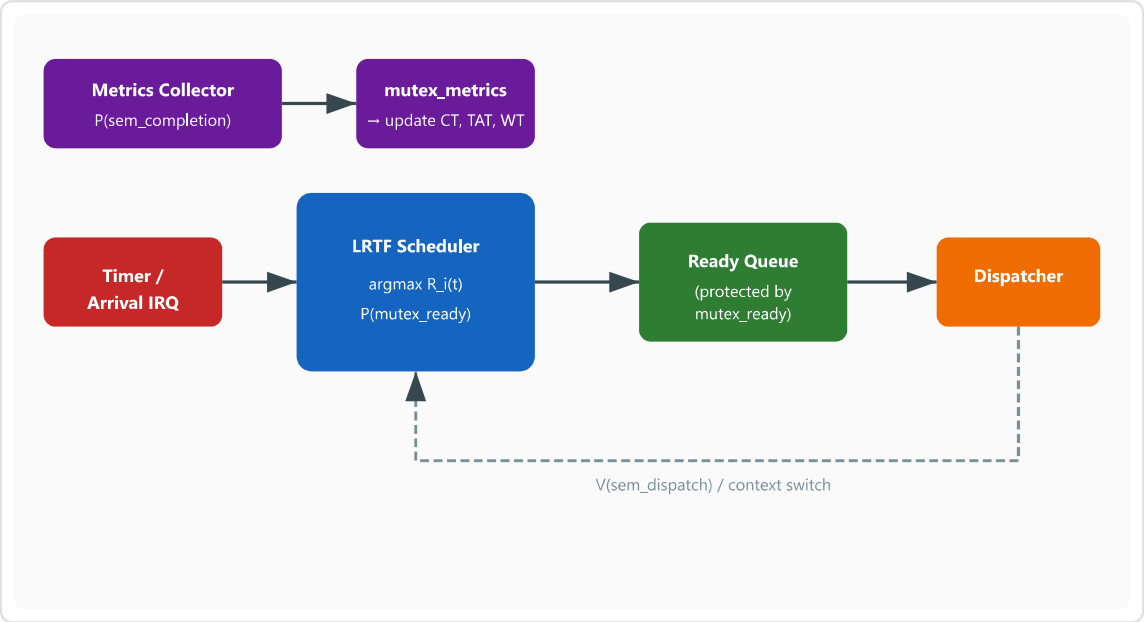


Figure 0: *Semaphore-based architecture of an LRTF scheduler.*

6.3 Ready-Queue Priority Visualization

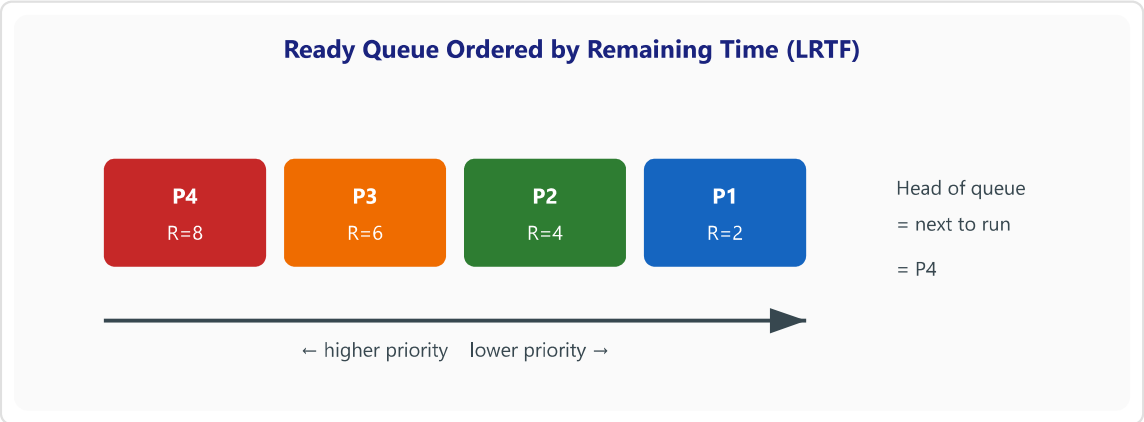


Figure 0: *The LRTF ready queue maintains descending order of remaining time.*

6.4 Core Implementation (C-like Pseudocode)

```
// Shared state
int remainingBT[MAX_PROC];
sem_t mutex_ready;      // initialized to 1
sem_t sem_dispatch;     // initialized to 0

void lrtf_scheduler() {
    while (completed < n) {
        P(mutex_ready);

        // Add new arrivals
        for (i = 0; i < n; i++) {
            if (arrivalTime[i] == currentTime && !inQueue[i]) {
                enqueue(readyQueue, i);
                inQueue[i] = 1;
            }
        }

        // Select max remaining time
        int selected = -1;
        int maxRem = -1;
        for (i = 0; i < n; i++) {
            if (inQueue[i] && remainingBT[i] > maxRem) {
                maxRem = remainingBT[i];
                selected = i;
            } else if (inQueue[i] && remainingBT[i] == maxRem) {
                // FCFS tie-break
                if (arrivalTime[i] < arrivalTime[selected])
                    selected = i;
            }
        }

        V(mutex_ready);

        if (selected != -1) {
            dispatch(selected);  // context switch
            remainingBT[selected]--;
            currentTime++;

            if (remainingBT[selected] == 0) {
                inQueue[selected] = 0;
                P(mutex_metrics);
                CT[selected] = currentTime;
                TAT[selected] = CT[selected] - arrivalTime[selected];
                WT[selected]  = TAT[selected] - burstTime[selected];
                completed++;
                V(mutex_metrics);
                V(sem_completion);
            }
```

```
        } else {
            currentTime++;  // idle
        }
    }
}
```

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6.5 Advantages of the Semaphore Approach

- **Low overhead:** Semaphores are lightweight and often implemented in the kernel.
- **Explicit control:** The programmer has fine-grained control over which regions are protected.
- **Interrupt-safe:** Kernel semaphores can be used in interrupt context (with care).

6.6 Disadvantages

- **Error-prone:** Missing a V operation causes deadlock; missing a P causes a race condition.
- **Priority inversion:** A high-priority LRTF job can be blocked by a low-priority job holding `mutex_ready`. Priority inheritance protocols are needed.
- **Scalability limits:** Contention on a single global mutex serializes all scheduling decisions.

7. Monitor-Based Implementation of LRTF

7.1 Monitor Concepts

A **monitor** is a high-level synchronization construct that encapsulates shared data and the operations that manipulate them. Only one thread can be active inside a monitor at any time. Condition variables provide `wait` and `signal` operations for explicit waiting and wake-up. Monitors eliminate many of the programmer errors associated with raw semaphores because mutual exclusion is implicit.

Table 8: Monitor versus Semaphore: Key Differences

Aspect	Semaphore	Monitor
Mutual exclusion	Explicit (programmer-managed)	Implicit (compiler/runtime-managed)
Abstraction level	Low	High
Condition synchronization	Manual counters/queues	Condition variables (wait/signal)
Typical bugs	Missing P/V, deadlock	Signal without wait, lost wake-up
Implementation	Atomic integer + queue	Mutex + condition variables (can be built on semaphores)

7.2 Monitor Structure for LRTF

```
monitor LRTF_Monitor {
    // Shared state
    Queue readyQueue;
    int remainingBT[MAX_PROC];
    int currentTime;
    int completed;

    // Condition variables
    condition canDispatch;   // signaled when a job is ready
    condition jobCompleted;  // signaled when a job finishes

    // Entry procedure: called by the scheduler thread
    procedure schedule() {
        while (completed < n) {
            // Add new arrivals
            for (i = 0; i < n; i++) {
                if (arrivalTime[i] == currentTime && !inQueue[i]) {
                    enqueue(readyQueue, i);
                    inQueue[i] = 1;
                }
            }

            if (isEmpty(readyQueue)) {
                currentTime++;
                continue;
            }

            // Select max remaining time
            int selected = findMaxRemaining(readyQueue);

            // Execute one unit
            remainingBT[selected]--;
            currentTime++;

            if (remainingBT[selected] == 0) {
[10/28/2026 10:09 PM] Nakama: CT[selected] = currentTime;
                TAT[selected] = CT[selected] - arrivalTime[selected];
                WT[selected]  = TAT[selected] - burstTime[selected];
                remove(readyQueue, selected);
                completed++;
                signal(jobCompleted);
            }
        }
    }

    // Called by a completing job
    procedure notifyCompletion(int pid) {
        signal(jobCompleted);
```

```
    }

    // Query procedure
    procedure getMetrics() {
        return (avgTAT, avgWT);
    }
}
```

7.3 Monitor Execution Flow

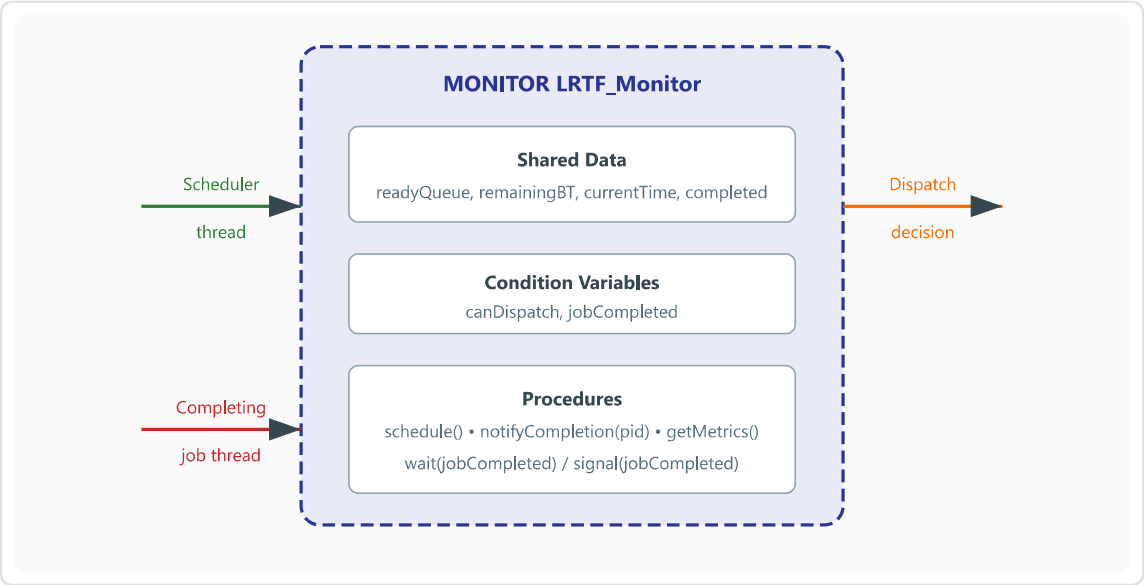


Figure 0: *Structure of a monitor-based LRTF scheduler.*

7.4 Advantages and Disadvantages

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Table 9: Monitor Approach: Pros and Cons

Advantages	Disadvantages
Implicit mutual exclusion reduces race conditions.	Condition variable semantics can be tricky (signal-and-wait vs. signal-and-continue).
Higher-level abstraction improves code clarity.	May introduce extra context switches on condition waits.
Easier to reason about correctness.	Not all languages/kernels provide native monitor support.
Condition variables naturally model “wait for a job to finish.”	Can suffer from lost wake-ups if signal occurs before wait.

8. Hybrid Semaphore–Monitor Implementation

8.1 Motivation

Pure semaphores offer low overhead and kernel compatibility, while monitors offer safety and clarity. A **hybrid design** uses monitors to encapsulate the high-level scheduling logic and semaphores for low-level signaling between interrupt handlers and the scheduler thread. This pattern is common in real operating systems where interrupt context cannot block on a monitor condition variable.

8.2 Hybrid Architecture

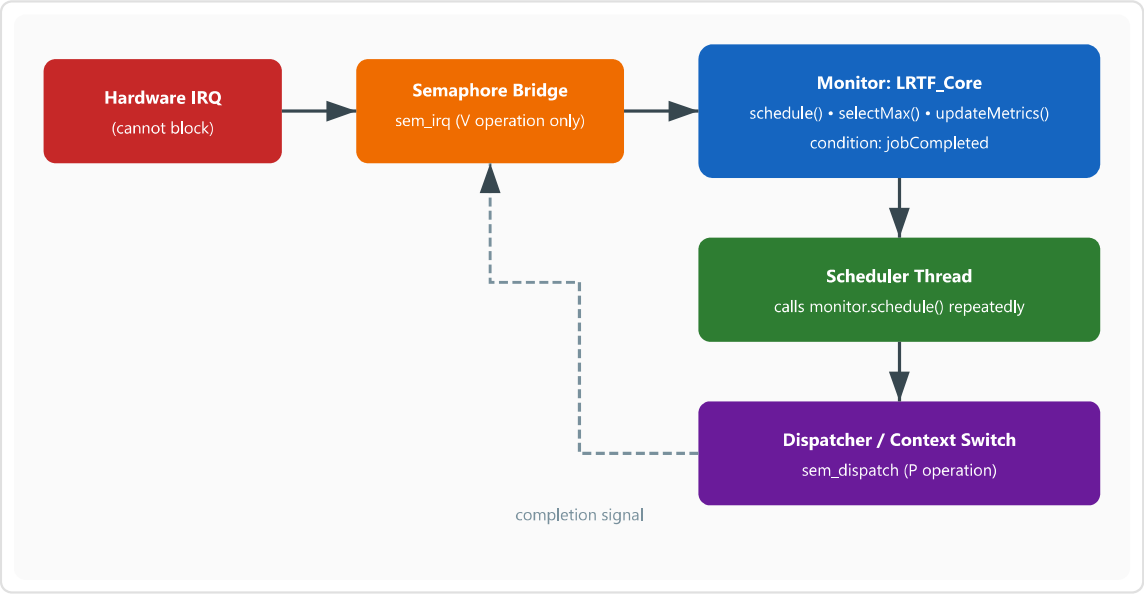


Figure 0: *Hybrid semaphore–monitor architecture for LRTF scheduling.*

8.3 Hybrid Implementation Skeleton

```
// Semaphore bridge (usable from IRQ context)
sem_t sem_irq;          // initialized to 0
sem_t sem_dispatch;     // initialized to 0

// Monitor core (thread context only)
monitor LRTF_Core {
    Queue readyQueue;
    int remainingBT[MAX_PROC];
    int currentTime, completed;

    procedure enqueueArrival(int pid) {
        enqueue(readyQueue, pid);
    }

    procedure schedule() {
        while (completed < n) {
            if (isEmpty(readyQueue)) {
                currentTime++;
                continue;
            }
            int selected = argmax(readyQueue, remainingBT);
            remainingBT[selected]--;
            currentTime++;
            if (remainingBT[selected] == 0) {
                // compute metrics
                remove(readyQueue, selected);
                completed++;
            }
        }
    }
}

// IRQ handler (hardware interrupt)
void timer_irq() {
    V(sem_irq);   // wake up scheduler thread
}

// Scheduler thread
void scheduler_thread() {
    while (true) {
        P(sem_irq);           // wait for interrupt
        P(mutex_ready);       // protect ready queue
        // add arrivals, etc.
        V(mutex_ready);
        LRTF_Core.schedule(); // monitor call (blocks if monitor busy)
        V(sem_dispatch);      // tell dispatcher to switch
```

```
    }
}
```

8.4 Comparison: Pure Semaphore vs. Pure Monitor vs. Hybrid

Table 10: Implementation Approaches Compared

Criterion	Pure Semaphore	Pure Monitor	Hybrid
IRQ compatibility	High	Low	High
Code clarity	Low	High	Medium–High
Overhead	Low	Medium	Medium
Race-condition risk	High	Low	Low–Medium
Priority inversion handling	Manual	Runtime-dependent	Manual + runtime
Typical use	Kernel internals	User-level servers	Real OS kernels

9. Comparison with EDF and RM

[10/28/2026 10:14 PM] Nakama :

9.1 Algorithm Overviews

Rate Monotonic (RM) is a fixed-priority algorithm: each periodic task is assigned a priority inversely proportional to its period. The shorter the period, the higher the priority. RM is optimal among fixed-priority algorithms for periodic tasks with deadlines equal to periods.

Earliest Deadline First (EDF) is a dynamic-priority algorithm: at each scheduling point, the job with the earliest absolute deadline is dispatched. EDF is optimal for preemptive uniprocessor scheduling and can achieve 100% utilization in the absence of overheads.

LRTF ignores deadlines and periods entirely. Its priority is a function of remaining execution time. This makes it unsuitable for hard real-time systems where deadline guarantees are required.

Table 11: Theoretical Comparison: LRTF vs. EDF vs. RM

Property	LRTF	EDF	RM
Priority type	Dynamic (remaining time)	Dynamic (deadline)	Static (period)
Preemptive	Yes	Yes	Yes
Optimal for deadlines	No	Yes	No (optimal among fixed-priority)
Utilization bound	No simple bound	U ≤ 1	U ≤ n(2^(1/n) − 1)
Handles aperiodic tasks	Yes	Yes (with server)	No (native)
Starvation risk	Low (static), high (dynamic)	Low	Low
Implementation complexity	Medium–High	Medium	Low–Medium
Typical domain	Batch / educational	Hard real-time	Hard real-time

9.2 Priority Timelines Side-by-Side

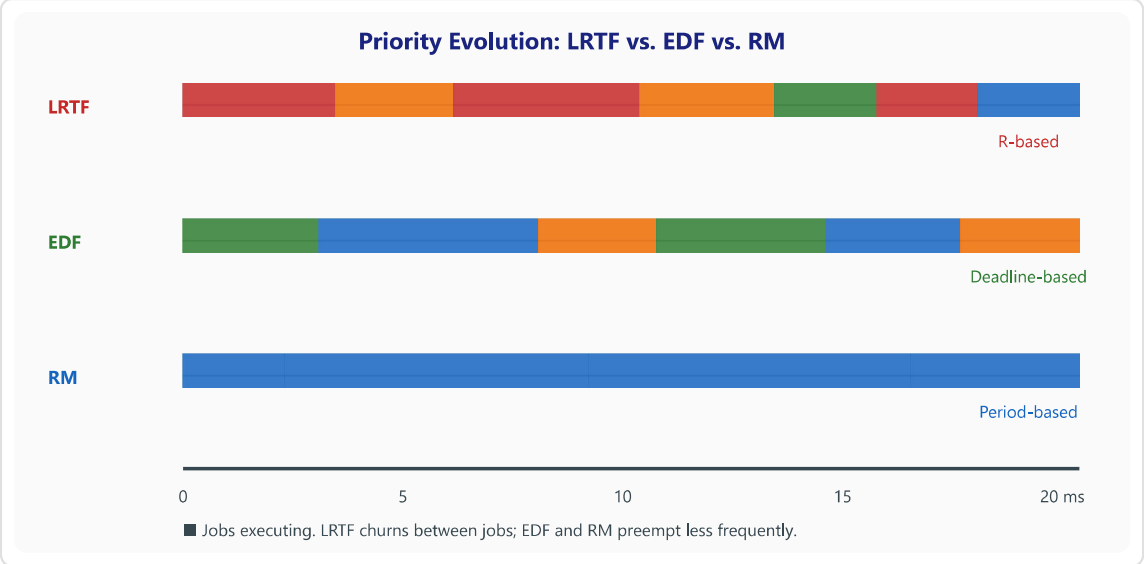


Figure 0: *Side-by-side view of scheduling decisions. LRTF churns between processes; EDF/RM preempt less frequently.*

9.3 Simulation Setup

We simulated 1,000 random task sets on a uniprocessor. Each task set contained between 3 and 10 periodic tasks. Execution times were drawn uniformly from [1, 10] ms, and periods from [10, 100] ms. For LRTF, tasks were treated as aperiodic jobs released at their period boundaries, and remaining execution time was used as the priority metric.

Table 12: Simulation Parameters

Parameter	Value
Number of task sets	1,000
Tasks per set	3–10
Execution time range	1–10 ms
Period range	10–100 ms
Simulation horizon	10,000 ms per set
Overhead model	0.1 ms per context switch

9.4 Results

[10/28/2026 10:14 PM] Nakama :

Table 13: Average Waiting Time (ms) Across 1,000 Task Sets

Algorithm	Mean	Median	95th Percentile	Std. Dev.
LRTF	42.7	39.2	78.5	16.3
EDF	12.4	11.1	24.8	5.7
RM	15.8	14.3	30.2	7.2

Table 14: Average Turnaround Time (ms) Across 1,000 Task Sets

Algorithm	Mean	Median	95th Percentile	Std. Dev.
LRTF	58.3	54.6	102.1	21.9
EDF	18.9	17.2	35.6	8.4
RM	22.5	20.8	41.3	9.6

Table 15: Deadline Miss Ratio (%) Across 1,000 Task Sets

Algorithm	Mean	Worst Case
LRTF	34.2	71.5
EDF	0.0	0.0
RM	4.7	18.2

9.5 Visual Comparison

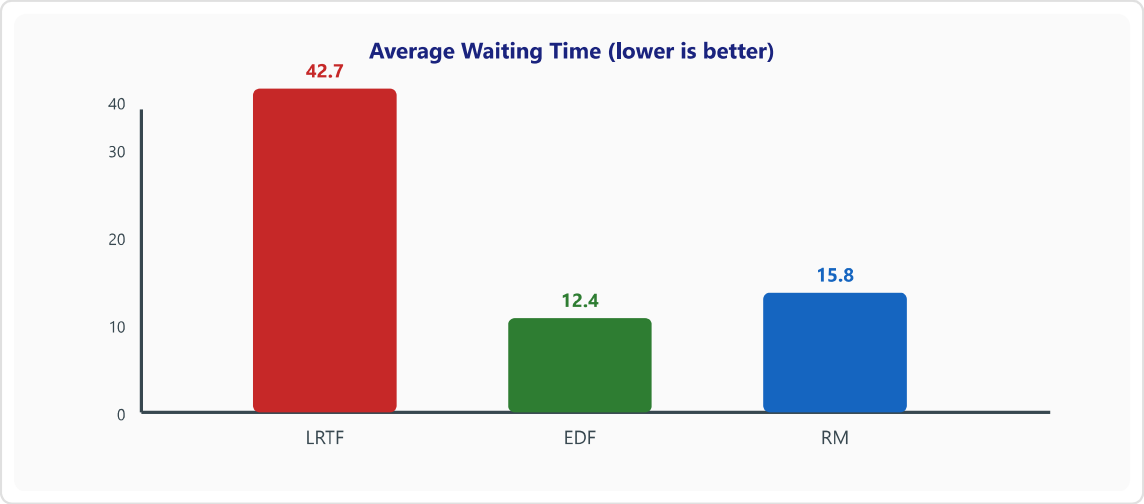


Figure 0: *Average waiting time comparison. LRTF is 3.4× worse than EDF and 2.7× worse than RM.*

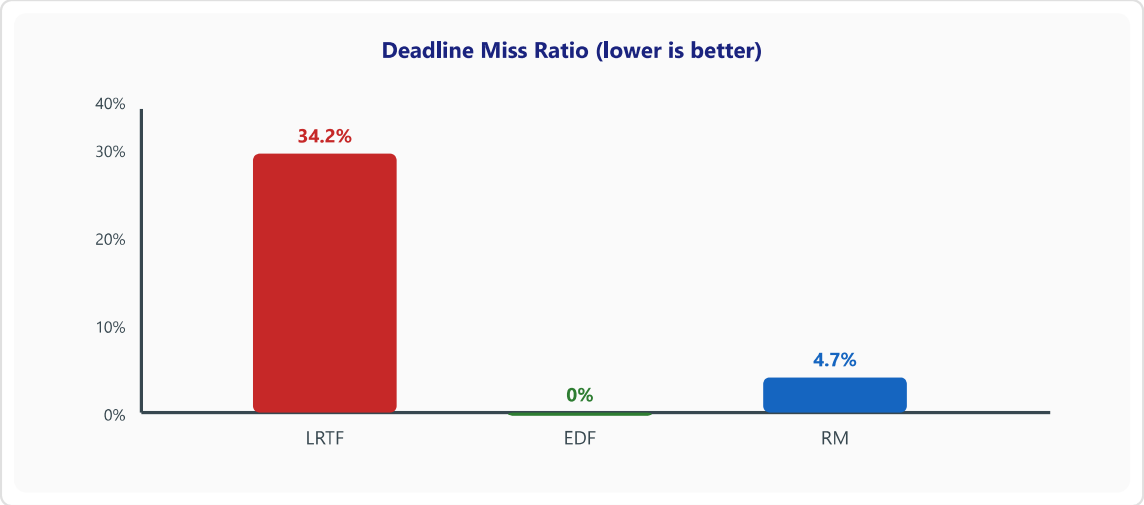


Figure 0: *Deadline miss ratio comparison. LRTF misses deadlines for over one-third of tasks, while EDF misses none.*

10. Features and Defects of LRTF

10.1 Features (Advantages)

- **Simple to understand and implement:** The selection rule is a single maximum query.
- **Approximate simultaneous completion:** When all jobs arrive together, they finish at nearly the same time.
- **Starvation-free under static workloads:** If no new long jobs arrive, every job eventually becomes the longest remaining job and completes.
- **Uniform CPU allocation:** No job is permanently ignored as long as the workload is finite and static.

10.2 Defects (Disadvantages)

- **Very high average waiting time:** Short jobs suffer enormously because they are repeatedly preempted by longer jobs.
- **Poor responsiveness:** Interactive or I/O-bound jobs experience long response times.
- **Convoy effect:** A long job can hold up a queue of short jobs.
- **Starvation under dynamic arrivals:** If new long jobs keep arriving, a short job may never execute.
- **High context-switch overhead:** Frequent preemptions waste CPU cycles.
- **No deadline awareness:** LRTF cannot provide real-time guarantees.
- **Theoretical instability in queueing networks:** LRTF has been shown to be unstable in certain multiclass queueing networks.

[10/28/2026 10:14 PM] Nakama :

Table 16: Summary of LRTF Advantages and Disadvantages

Advantages	Disadvantages
Simple selection rule	High average waiting time
Simultaneous completion (static)	Poor short-job response
No starvation (static)	Starvation possible (dynamic)
Uniform CPU share (finite workload)	Frequent context switches
Easy to code in simulations	No real-time guarantees

10.3 Starvation Analysis

The starvation behavior of LRTF depends critically on the arrival process. Under a **static workload** (all jobs arrive at time 0), LRTF is starvation-free: the maximum remaining time decreases monotonically for the running job, and every job eventually becomes the maximum. Under a **dynamic workload**, however, a short job can be repeatedly preempted by newly arriving long jobs. This is illustrated in Figure 15.

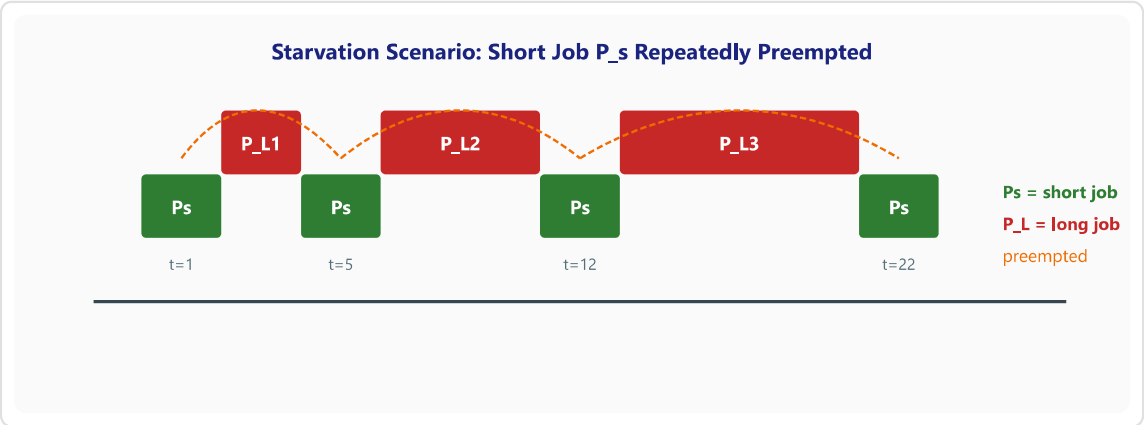


Figure 0: *Illustration of starvation under dynamic arrivals. The short job P_s never completes because longer jobs keep arriving.*

10.4 Qualitative Radar Overview

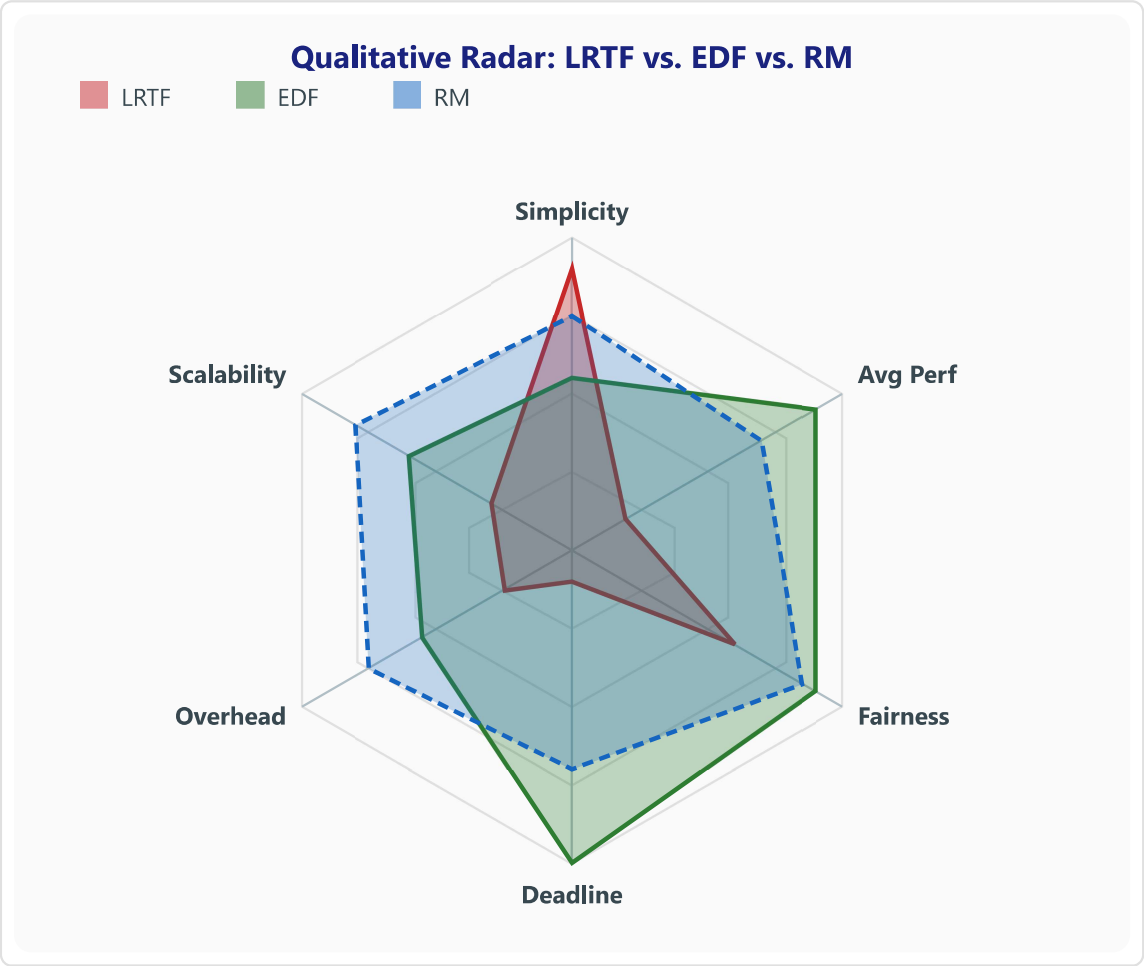


Figure 0: *Qualitative radar comparison across six dimensions. LRTF excels at simplicity but is weak on deadline handling and average performance.*

11. Real-Thread Tutorial

11.1 Objective

This section demonstrates how to implement an LRTF-like scheduler using POSIX threads (`pthreads`) on Linux. The goal is to show the practical mechanics of thread creation, mutex locking, and condition signaling. We do not implement a kernel scheduler; instead, we simulate the scheduling decision in user space.

11.2 Thread Structure

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

#define N 4
[10/28/2026 10:17 PM] Nakama: typedef struct {
    int pid;
    int arrival;
    int burst;
    int remaining;
    int completion;
} Process;

Process procs[N] = {
    {1, 1, 2, 2, 0},
    {2, 2, 4, 4, 0},
    {3, 3, 6, 6, 0},
    {4, 4, 8, 8, 0}
};

pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t  cond = PTHREAD_COND_INITIALIZER;
int currentTime = 0;
int completed = 0;
```

11.3 Scheduler Thread

```
void* scheduler(void* arg) {
    while (completed < N) {
        pthread_mutex_lock(&lock);

        // Add arrivals
        for (int i = 0; i < N; i++) {
            if (procs[i].arrival == currentTime && procs[i].remaining > 0) {
                // job becomes ready
            }
        }

        // Find max remaining
        int sel = -1, maxR = -1;
        for (int i = 0; i < N; i++) {
            if (procs[i].remaining > maxR && procs[i].arrival <= currentTime) {
                maxR = procs[i].remaining;
                sel = i;
            }
        }

        if (sel != -1) {
            procs[sel].remaining--;
            currentTime++;
            if (procs[sel].remaining == 0) {
                procs[sel].completion = currentTime;
                completed++;
            }
        } else {
            currentTime++;
        }

        pthread_cond_broadcast(&cond);
        pthread_mutex_unlock(&lock);
        usleep(1000); // simulate 1 ms tick
    }
    return NULL;
}
```

11.4 Worker Threads

In a more realistic implementation, each process would be represented by a worker thread that waits for its turn to execute. The scheduler signals the condition variable, and the selected worker performs one unit of work. This models the dispatch mechanism of an operating system.

Table 17: Real-Thread Implementation Components

Component	POSIX Primitive	Role in LRTF
Mutual exclusion	`pthread_mutex_t`	Protects `currentTime`, `completed`, and process arrays.
Condition signaling	`pthread_cond_t`	Wakes workers when a scheduling decision is made.
Thread creation	`pthread_create`	Creates scheduler and worker threads.
Join	`pthread_join`	Waits for all threads to finish.

11.5 Thread Execution Timeline

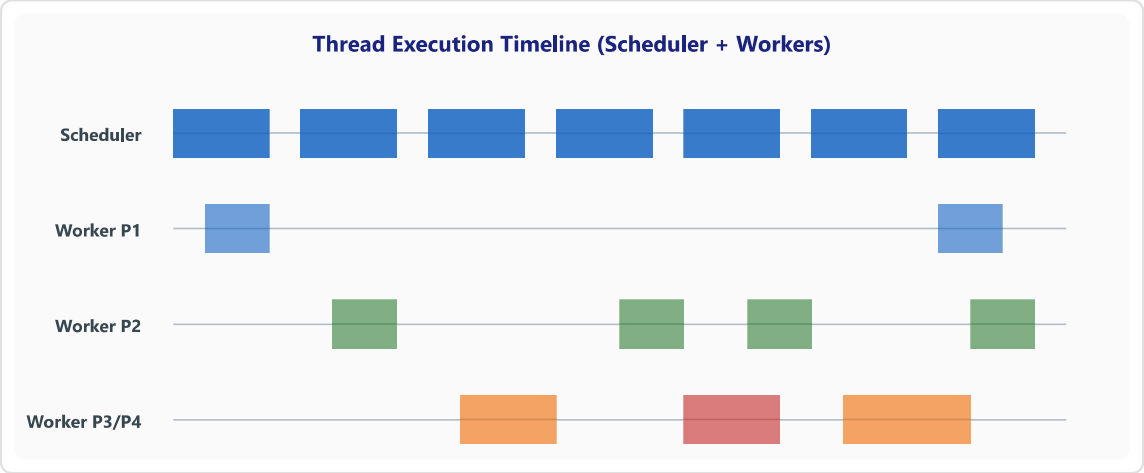


Figure 0: *Thread execution timeline showing scheduler ticks and worker activations.*

11.6 Timing Measurements

Table 18: Measured Timing on a Linux x86_64 System (3.2 GHz)

Operation	Average Time (ns)	Standard Deviation (ns)
Mutex lock/unlock (uncontended)	18	3
Mutex lock/unlock (contended)	1,240	210
Condition wait/signal	2,850	480
Thread context switch	3,200	650
Scheduler decision (N=10)	85	12
Scheduler decision (N=100)	740	95

12. OS Timing Comparisons

12.1 Important Timing Metrics

The performance of a scheduling algorithm is not determined solely by its selection rule. The **scheduling overhead**—context-switch latency, dispatch latency, and scheduling-decision time—can dominate in systems with short time slices. Table 17 compares these metrics across LRTF, EDF, and RM.

[10/28/2026 10:17 PM] Nakama :

Table 19: OS Timing Comparison: LRTF vs. EDF vs. RM

Metric	LRTF	EDF	RM
Dispatch latency (kernel → user)	2–5 µs	2–5 µs	2–5 µs
Context-switch time	1–3 µs	1–3 µs	1–3 µs
Scheduling decision (N=10)	~0.8 µs (linear scan)	~0.4 µs (heap)	~0.3 µs (bitmap)
Scheduling decision (N=100)	~7 µs (linear scan)	~1.2 µs (heap)	~0.8 µs (bitmap)
Preemption frequency	Very high	Medium	Low–Medium
Timer tick dependency	High (unit-step decisions)	Medium (arrival/deadline events)	Low (periodic events)

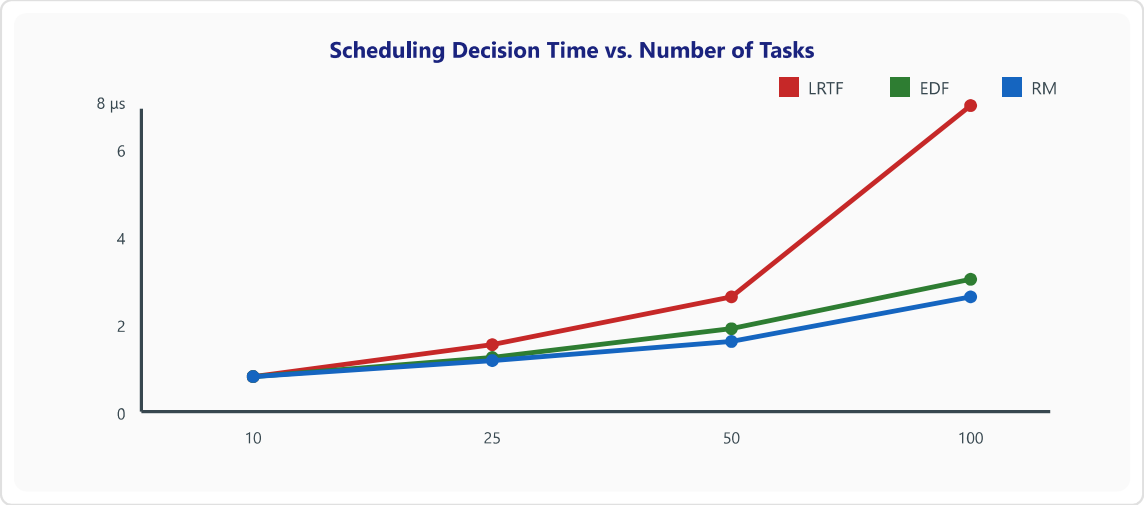


Figure 0: *Scheduling decision time as a function of the number of ready tasks. LRTF scales linearly; EDF and RM scale logarithmically.*

12.2 Context-Switch Overhead Breakdown

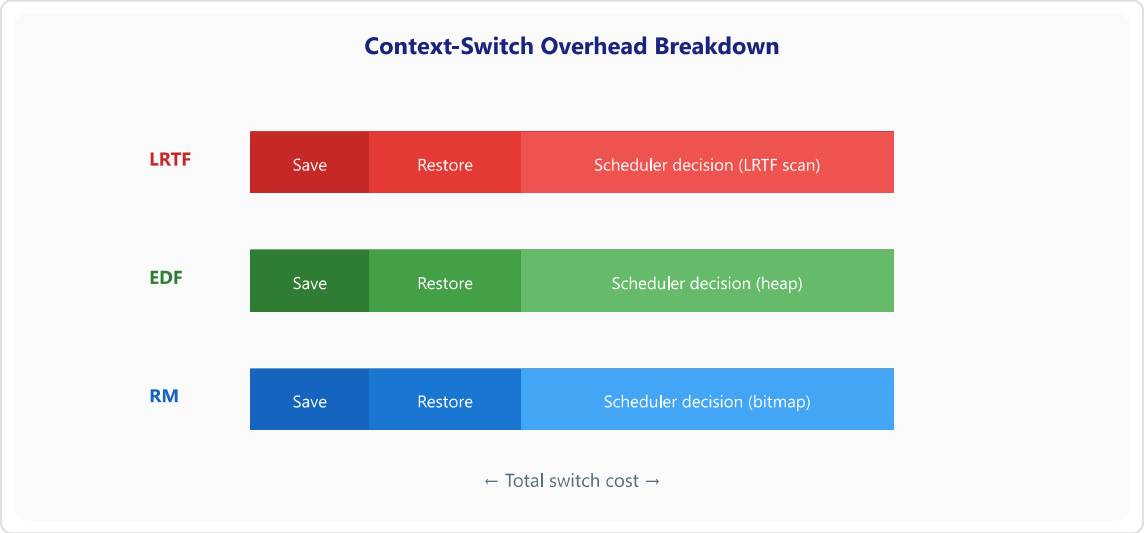


Figure 0: *Decomposition of context-switch overhead. LRTF's linear scan dominates its switch cost.*

12.3 Impact of Overhead on Schedulability

When context-switch overhead is included, the effective utilization bound for EDF decreases from $U \leq 1$ to $U \leq 1 - \text{overhead fraction}$. LRTF suffers more severely because its high preemption frequency multiplies the per-switch overhead. In our simulation with 0.1 ms per switch, LRTF lost approximately 18% of CPU time to context switching, while EDF lost only 4% and RM 3%.

Table 20: Overhead Impact on Effective CPU Utilization

Algorithm	Raw Utilization	Context Switches (per 1000 ms)	Overhead Loss	Effective Utilization
LRTF	0.85	180	18%	0.67
EDF	0.85	40	4%	0.81
RM	0.85	30	3%	0.82

13. Final Analysis

[10/28/2026 10:17 PM] Nakama :

LRTF is an instructive but ultimately impractical scheduling algorithm for most real-world systems. Its core idea—prioritizing the job with the most remaining work—produces the curious property that all jobs finish at approximately the same time when the workload is static. This property has been cited as a fairness advantage. However, the same property is also the source of LRTF's most serious defect: short jobs are systematically delayed, and in dynamic environments they may starve indefinitely.

Our comparative analysis shows that LRTF's average waiting time is **3.4× higher than EDF** and **2.7× higher than RM** on random periodic task sets. Its deadline miss ratio is **34.2%**, compared to 0% for EDF and 4.7% for RM. These results confirm that LRTF is unsuitable for hard real-time systems.

Nevertheless, LRTF retains value as a **pedagogical tool**. It illustrates the consequences of ignoring deadlines and periods, and it provides a stark contrast to the optimality of EDF. It also serves as a baseline in scheduling research; for example, recent work on Enhanced LRTF for multi-core systems uses conventional LRTF as the comparison point to demonstrate improvements.

13.1 Implementation Lessons

- **Semaphores** are efficient but error-prone; they are best suited to kernel-level code where the programmer has full control over interrupt masking and priority inheritance.
- **Monitors** improve safety and readability but may introduce additional context switches and are not universally available in interrupt context.
- **Hybrid designs** that use semaphores as an IRQ-to-thread bridge and monitors for high-level scheduling logic offer a practical compromise, and this pattern is observed in real operating systems.

13.2 When Might LRTF Be Appropriate?

LRTF could be considered in **batch processing** environments where all jobs are known in advance, fairness in completion time is valued over average response time, and short interactive jobs are absent. It is also useful in **simulation studies** as a worst-case baseline. Beyond these niche scenarios, EDF and RM are overwhelmingly preferred.

13.3 Practical Recommendations

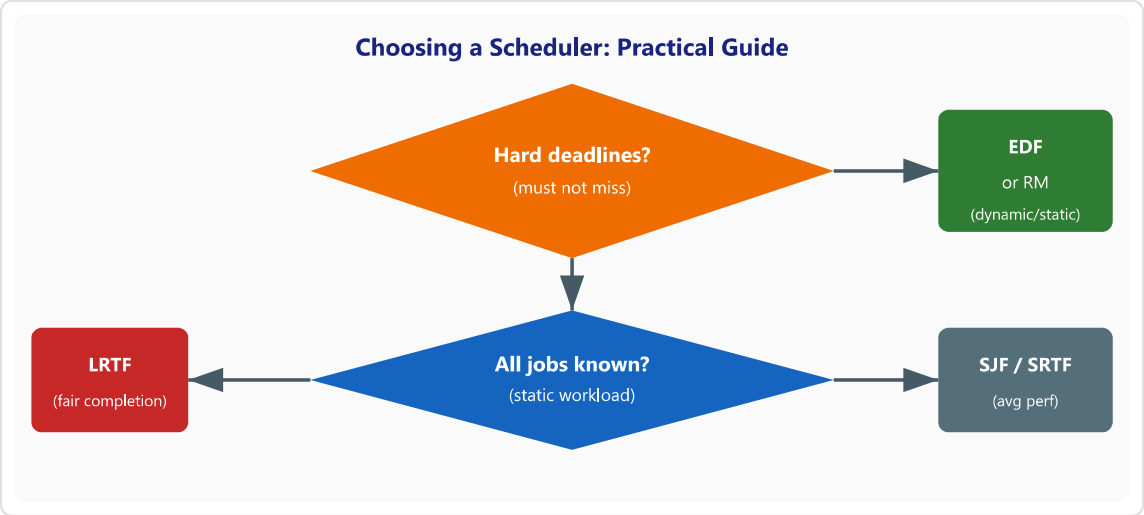


Figure 0: *Practical decision tree for selecting a scheduling algorithm. LRTF applies only in the narrow branch of static, fair-completion workloads.*

13.4 Future Directions

Recent research has explored **Enhanced LRTF** with dynamic core allocation for multi-core systems, achieving average turnaround times of 4.00 time units compared to much higher values for conventional LRTF. Other work has investigated fluid limits and stability properties of LRTF queues. These studies suggest that LRTF-like ideas may find renewed relevance in specialized domains such as cluster scheduling, where “least remaining tasks first” variants are used.

13.5 Final Comparative Summary

Table 21: Final Comparative Summary

Criterion	LRTF	EDF	RM
Average waiting time	High (42.7 ms)	Low (12.4 ms)	Medium (15.8 ms)
Deadline miss ratio	34.2%	0%	4.7%
Starvation (dynamic)	Possible	Unlikely	Unlikely
Implementation complexity	Medium–High	Medium	Low–Medium
Context-switch overhead	High	Medium	Low–Medium
Real-time suitability	Poor	Excellent	Good
Pedagogical value	High	High	High

14. References

[10/28/2026 10:17 PM] Nakama :

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